Commutative Algebra Problem Set #4

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Problem (Exercise 1)**.** *Let A be a domain and let M be a finitely presented A-module. Prove that M is flat (equivalently, projective or locally free) if and only if the function*

$$\delta_M : Spec(A) \to \mathbb{N}, \quad \delta_M(x) = \dim_{\kappa(x)} M(x)$$

is constant. Find a counterexample to the "if" part with A not a domain.

Solution. ($\Rightarrow$). Assume M is flat and f.p. (and thus locally free), then localization $M_x \cong A_x^{n_x}$ for every $x \in \mathrm{Spec}(A)$ and some $n_x \in \mathbb{Z}_{\geq 0}$. Let us find $\delta_M(x)$:

$$\begin{aligned}\delta_M(x) &= \dim_{\kappa(x)} M(x) = \dim_{\kappa(x)}(M \otimes_A \kappa(x)) \\ &= \dim_{\kappa(x)}(M_x \otimes_{A_x} \kappa(x)) \\ &= \dim_{\kappa(x)}(A_x^{n_x} \otimes_{A_x} \kappa(x)) \\ &= \dim_{\kappa(x)}(\kappa(x)^{n_x}) = n_x.\end{aligned}$$

Thus we shall show that $M_x \cong A_x^n$ for some constant $n \in \mathbb{Z}_{\geq 0}$ not depending on $x \in \mathrm{Spec}(A)$. Let $K := \mathrm{Frac}(A)$. We claim that $n = \dim_K M_K$. Indeed, note that $M_K \cong (M_x) \otimes_{A_x} K \cong K^{n_x}$. By taking $\dim_K(\cdot)$ of both sides, we get $n_x = n$, which completes this part.

($\Leftarrow$). Now assume $\delta_M(x) = n$ for all $x \in \mathrm{Spec}(A)$. We shall show that M is locally free. As noted in the hint, replace A with the local ring A_x and M with M_x for arbitrary $x \in \mathrm{Spec}(A)$. Note that the function δ_{M_x} is also trivially constant.

Let us drop the subscript x and assume that $(A, \mathfrak{m})$ is a local domain with maximal ideal $\mathfrak{m} = s \in \mathrm{Spec}(A)$. Thus, we have

$$\delta_M(s) = \dim_{A/\mathfrak{m}}(M \otimes_A A/\mathfrak{m}) = \dim_{A/\mathfrak{m}}(M/\mathfrak{m}M) = n.$$

Using Nakayama's lemma, we infer that M is generated by n elements, thus defining the following exact sequence:

$$0 \to N \to A^n \xrightarrow{f} M \to 0, \quad N = \ker(f)$$

Now apply $(\cdot) \otimes_A K$ functor (and it is exact because K is a flat A-module since it is a localization of A):

$$0 \to N \otimes_A K \to K^n \xrightarrow{f \otimes 1} M \otimes_A K \to 0$$

Now note that $\dim_K(M \otimes_A K) = n$ and $f \otimes 1$ is a surjective linear map from K^n to a K-vector space of dimension n. This implies that $f \otimes 1$ is an isomorphism and thus $N \otimes_A K = 0$. This implies that $N = 0$ and thus original map $f : A^n \to M$ is an isomorphism. Thus, $M \cong A^n$, so M is free.

Now recall that by M we actually meant M_x, so we got $M_x \cong (A_x)^n$. We conclude that M is locally free. □

Counterexample. Let us construct a ring A that is not a domain, a finitely presented A-module M, such that $\delta_M(x)$ is constant, but M is not flat.

Take $A := k[T]/(T^2)$ and $M := k = A/(T)$. A is clearly not a domain and M is finitely presented. Now note that $\operatorname{Spec}(A) = \{\mathfrak{m}\}$ for $\mathfrak{m} = (T)$, so the function δ_M has only one input:

$$\delta_M(\mathfrak{m}) = \dim_{\kappa(\mathfrak{m})}(M \otimes_A \kappa(\mathfrak{m})) = \dim_k(k \otimes_A k) = 1.$$

Let us show that M is not flat. We need to find some ideal I such that $I \hookrightarrow A$ is injective but $\sigma : M \otimes_A I \to M \otimes_A A$ is not. Indeed, simply take $I := (T)$. Take non-zero element $z := 1 \otimes T \in M \otimes_A I$. σ sends this element to $1 \otimes T$, but in $M \otimes_A A$. But inside $M \otimes_A A$, $1 \otimes T = T \otimes 1$, which is 0.

Problem (Exercise 2)**.** *Prove that the following ring homomorphisms are not flat.*

(a) $k[x^2, x^3] \hookrightarrow k[x]$;

(b) $k[x^2, xy, y^2] \hookrightarrow k[x, y]$;

(c) $k[x, xy] \hookrightarrow k[x, y]$.

Solution. For all parts for $A \hookrightarrow B$ we shall show that the functor $(\cdot) \otimes_A B$ is not left exact, that is for some ideal $I \subseteq A$

$$\{0 \to I \to A \text{ exact}\} \text{ such that } \{0 \to I \otimes_A B \to B\} \text{ is not exact,}$$

where the map $\sigma : I \otimes_A B \to B$ is given by $i \otimes b \mapsto ib$.

Part (a). Take $I := (x^2, x^3)$. We claim that

$$z := x^2 \otimes x - x^3 \otimes 1 \in \ker(\sigma) \setminus \{0\}.$$

Clearly $z \in \ker(\sigma)$ since $\sigma(z) = x^3 - x^3 = 0$. However, I honestly cannot rigorously show that z is non-zero. Intuition is the following: z were zero if we could "factor out" x somehow. However, *x is not an element of $k[x^2, x^3]$*, thus this operation is impossible. There is no other way how we can get 0 and thus this element is non-zero.

Part (b). Take $I := (x^2, xy)$. Then,

$$z := x^2 \otimes y - xy \otimes x \in \ker(\sigma) \setminus \{0\}.$$

Part (c). Take $I := (x, xy)$. Then,

$$z := x \otimes y - xy \otimes 1 \in \ker(\sigma) \setminus \{0\}.$$

Problem (Exercise 3)**.** *Let A be an $\mathbb{F}_p$-algebra and define the **Frobenius morphism** of A as $\pi : A \to A, x \mapsto x^p$. Prove that π induces the identity map $Spec(A) \to Spec(A)$.*

Proof. By definition, a ring homomorphism $f : A \to B$ induces a continuous map $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ defined by $f^*(\mathfrak{q}) = f^{-1}(\mathfrak{q})$ for any $\mathfrak{q} \in \mathrm{Spec}(B)$. In our case, we have a ring homomorphism $\pi : x \mapsto x^p$, and we shall show that $\pi^*(\mathfrak{p}) = \mathfrak{p}$ for all $\mathfrak{p} \in \mathrm{Spec}(A)$.

Note that by definition $\pi^*(\mathfrak{p}) = \pi^{-1}(\mathfrak{p}) = \{a \in A : a^p \in \mathfrak{p}\}$. But note that $\mathfrak{p}$ is prime, so $a^p \in \mathfrak{p}$ automatically implies $a \in \mathfrak{p}$. The reverse implication is trivial. Thus, $\pi^*(\mathfrak{p}) = \mathfrak{p}$. □

Problem (Exercise 4)**.** *This problem consists of two parts:*

(a) Let $A = \mathbb{F}_p[T_1, \dots, T_n]$. Prove that π is finite and flat.

(b) Let $A = \mathbb{F}_p[T^2, T^3]$. Prove that π is finite but not flat.

Solution.

Part (a). Let us show that $A = \mathbb{F}_p[T_1, \dots, T_n]$ is a finite and flat $\mathbb{F}_p[T_1^p, \dots, T_n^p]$-module. Notice that it is finite. Indeed, consider arbitrary monomial $T_1^{\alpha_1} \dots T_n^{\alpha_n}$. Divide each α_i by p: $\alpha_i = p\beta_i + \gamma_i$ for $0 \le \gamma_i < p$. Thus:

$$T_1^{\alpha_1} \dots T_n^{\alpha_n} = (T_1^{\beta_1} \dots T_n^{\beta_n})^p \cdot (T_1^{\gamma_1} \dots T_n^{\gamma_n})$$

Note that $(T_1^{\beta_1} \dots T_n^{\beta_n})^p \in \mathbb{F}_p[T_1^p, \dots, T_n^p]$, and the remaining term is in

$$S := \{T_1^{\gamma_1} \dots T_n^{\gamma_n} : 0 \le \gamma_i < p \text{ for all } i\}$$

Thus, S generates A as an A'-module. Since $\#S = p^n$, π is finite. Moreover, A is a free A'-module since S forms a basis (trivially, all elements of S are linearly independent over $\mathbb{F}_p[T_1^p, \dots, T_n^p]$), so in particular π is flat.

Part (b). The proof of π being finite is the same as for *part (a)*, where the set S is given by:

$$S = \{T^{2i}T^{3j} : 0 \le i, j < p\}.$$

However, in this case S does not form a basis. As in Exercise 2, we shall find ideal $I \subseteq \mathbb{F}_p[T^{2p}, T^{3p}]$ such that $\sigma : I \otimes \mathbb{F}_p[T^2, T^3] \to \mathbb{F}_p[T^2, T^3]$ given by $a(T^p) \otimes b(T) \mapsto a(T^p)b(T)$ is not injective. Indeed, take $I = (T^{2p}, T^{3p})$. Notice that $z := T^{2p} \otimes (-T^p) + T^{3p} \otimes 1 \in \ker(\sigma) \setminus \{0\}$. □

Problem (Exercise 5). *Let* $A = k[x, y, z]/(x^2 + y^2 + z^2 - 1)$ *where* $k \not\supseteq \mathbb{F}_2$. *Consider the module of Kähler differentials* $M = \Omega^1_{A/k}$. *Show that* M *is locally free (or equivalently projective) and describe it as a direct summand of* A^3.

Solution. Recall that for $S = k[x_1, \dots, x_n]$, ideal $I = (f_1, \dots, f_r) \subseteq S$, and $A = S/I$, we have

$$\Omega_{A/k} \cong \frac{\Omega_{S/k}}{I\Omega_{S/k} + S(d(i) : i \in I)}.$$

We have $S = k[x, y, z]$ and $I = (\Gamma)$ for $\Gamma(x, y, z) = x^2 + y^2 + z^2 - 1$. Thus:

$$\Omega_{A/k} \cong \frac{\Omega_S}{\Gamma\Omega_S + Sd\Gamma} \cong \frac{\Omega_S/\Gamma\Omega_S}{(\Gamma\Omega_S + Sd\Gamma)/(\Gamma\Omega_S)} \cong \frac{A^3}{Ad\Gamma},$$

where $d\Gamma = 2xdx + 2ydy + 2zdz$. So let us develop some intuition of what we got so far. Note that element is zero in $\Omega_{A/k}$ iff it is of form $f(2xdx+2ydy+2zdz)$, which intuitively are vectors, normal to the sphere at (x, y, z) (scaled by some function $f \in A$). Thus, we expect $\Omega_{A/k} \cong A^3/d\Gamma$ to be the module of all tangent vector fields on the sphere.

Namely, suppose $\mathbf{u} = (u_x, u_y, u_z) \in A^3$ be any vector field and let $\boldsymbol{\nu} = (x, y, z)$ be the normal vector. Define the natural scalar product of $\mathbf{u}$ and $\boldsymbol{\nu}$ as $s : A^3 \to A$. Then, the projection of $\mathbf{u}$ onto $\boldsymbol{\nu}$ is $\pi : A^3 \to A^3$, $\mathbf{u} \mapsto s(\mathbf{u})\boldsymbol{\nu}$. Similarly, define the tangent component as $\tau : A^3 \to A^3$ as $\mathbf{u} \mapsto \mathbf{u} - s(\mathbf{u})\boldsymbol{\nu}$. Thus we see:

$$\mathbf{u} = \pi(\mathbf{u}) + \tau(\mathbf{u})$$

Now, suppose $N = \mathrm{im}(\pi)$ and $T := \mathrm{im}(\tau)$. Via the construction above, we have shown that $A^3 = N + T$. But clearly, $N \cap T = \{0\}$, so we have $A^3 = N \oplus T$. But notice that $N = \{f\boldsymbol{\nu} : f \in A\} \cong A$ and $T = \mathrm{im}(\tau) = \ker(\pi)$. Thus:

$$\Omega_{A/k} \cong \frac{A^3}{\mathrm{im}(\pi)} \cong \ker(\pi) \text{ and thus } \boxed{A^3 = A \oplus \Omega_{A/k}}$$

This shows that $\Omega_{A/k}$ is projective. □